

Analytical Problem Solving

- 38) convex Hull - Regular boundary with interior points
 Apply the Brute force convex hull algorithm on the set $\{(0,0), (0,5), (5,5), (5,0), (2,2), (3,3), (1,4)\}$. clearly interpret the Problem the Problem and explain the concept of determining the outer boundary of a point set. check all possibilities line pairs and identify which edges from the convex Hull show the logical steps used to exclude interior points from the boundary count total comparisons performed Analyze the complexity and interpret the practical significance of the result

39) Given set of points

$$S = \{(0,0), (0,5), (5,5), (5,0), (2,2), (3,3), (1,4)\}$$

Apply the Brute force convex Hull algorithm to determine the outer boundary of the given points

why Brute force

- It checks every possible pair of points
- For each pair it verifies whether all remaining points lie on the same side of line
- if yes that the line segment is convex hull edge
- suitable for small data sets

Given points

Points	coordinates		
A	(0,0)	E	(2,2)
B	(0,5)	F	(3,3)
C	(5,5)	G	(1,4)
D	(5,0)		

checking important line pairs

Edge AB : $(0,0) - (0,5)$

All remaining points lies on the right of line

AB is convex hull edge

Edge BC : $(0,5) - (5,5)$

All remaining points lies to the below line

BC is convex hull point

Edge CD : $(5,5) - (5,0)$

All remaining points lies to left of the line

CD is convex hull point

Edge DA : $(5,0) - (0,0)$

All remaining points be above the line

DA is convex hull point

Interior Point

The points

$(2,2)$, $(3,3)$, $(1,4)$

lie completely inside the rectangle

when ever they are tested with other line pairs
points exist on both sides of line so these pairs
cannot become hull edges
therefore these points are excluded

Convex hull

The convex hull consists of rectangle .

$(0,0) \rightarrow (0,5) \rightarrow (5,5) \rightarrow (5,0) \rightarrow (0,0)$

Number of comparisons .

$n = 7$ points

No of line pairs = $\frac{7 \times 6}{2} = 21$

Q5 Linear search - negative number Detection
 Given the unsorted dataset $[12, -7, 25, 39, -14, 48, 56]$
 apply linear search to find the element -14 . clearly
 interpret the problem and explain the search process
 step by step count the number of comparisons - required
 before locating the target. Analyze the best case, average case
 and worst case time complexity. Verify the correctness
 of the result. Interpret efficiency of linear search
 when searching for negative values in an unsorted list

Q7 Linear Search - negative number Detection

Given

$A = [12, -7, 25, 39, -14, 48, 56]$

find the element -14 using linear search

Interpretation

Linear search checks each element one by one from the beginning until the required element is found since the dataset is unsorted.

STEP by STEP search

STEP	Element	Comparison	Result
1	12	$12 \neq -14$	Continue
2	-7	$-7 \neq -14$	Continue
3	25	$25 \neq -14$	Continue
4	39	$39 \neq -14$	Continue
5	-14	$-14 = -14$	Found

Each pair is checked against the remaining

7-2=5
points

Total comparison

$$21 \times 5 = 100$$

$$\text{Total} = 105$$

Time complexity

For n points

$$\text{line pairs} = O(n^2)$$

$$\text{checking remaining points} = O(n)$$

Overall

$$O(n^3)$$

Verification

The four corner points.

(0,0)

(0,5)

(5,5)

(5,0)

From the smallest convex polygon enclosing all other points

The interior points

(2,2), (3,3), (1,4)

do not lie on the boundary and are therefore excluded.

therefore convex hull

69) Linear search - unsuccessful search in unsorted data

Given the unsorted dataset [16, 28, 39, 47, 58, 69, 74] apply linear search to find the element 100. Clearly interpret the problem and explain the sequential search process in detail. Count the total number of comparisons made before concluding that the element is absent. Analyze the best case, avg case, and worst case time complexity. Verify the correctness of the result. Interpret the efficiency of linear search is unsuccessful case.

Linear search

Given unsorted dataset

$A = [16, 28, 39, 47, 58, 69, 74]$

find the element 100 using linear search

Interpretation of the Problem

Linear search examines each element of the list sequentially from the first element until last element target element found after checking all element.

Step by Step search.

Step	Element	Comparison	Result
1	16	$16 \neq 100$	Continue
		$28 \neq 100$	continue
2	28	$39 \neq 100$	continue
3	39	$47 \neq 100$	continue
4	47	$58 \neq 100$	continue
5	58	$69 \neq 100$	continue
6	69	$74 \neq 100$	end
7	74		

Number of comparisons

To search compares target with

12

-7

25

39

-14

Total comparisons = 5

Time complexity

Best case: $O(1)$ element found at the first position

Average case: $O(n)$ element found in the middle

worst case: $O(n)$ element found at last position or
not present

Verification

The target element -14 is found at index 4
5th position

Therefore the linear search correctly locates
the required element

Linear search is simple and effective for
unsorted datasets because it does not require sorting
it searches negative and positive values in
same way. It becomes less efficient since it
may need to examine every element before
finding the required value.

Number of comparisons

The target is compared with every element

(1) 16

(2) 28

(3) 39

(4) 47

(5) 58

(6) 69

(7) 74

Total comparison 27

Time complexity

Best case $O(1)$ - The element found at first position

Average case $O(n)$ - The element is somewhere found in the middle or after several comparisons

Worst case :.

The element is at last position or is not present requiring all element to be checked.

Verification

The data set contains elements [16, 28, 39, 47, 58, 69, and 74] since 100 is not present the algorithm correctly reports that the element is absent for checking all seven elements.